SIMRAD

PRO SERIES

IS40 PRO
Operator Manual

ENGLISH

PRO.SIMRAD-YACHTING.COM

Preface

Disclaimer

As Navico is continuously improving this product, we retain the right to make changes to the product at any time which may not be reflected in this version of the manual. Contact your nearest distributor if you require any further assistance.

It is the owner's sole responsibility to install and use the instrument in a manner that will not cause accidents, personal injury or property damage. The user of this product is solely responsible for observing safe boating practices.

NAVICO HOLDING AS AND ITS SUBSIDIARIES, BRANCHES AND AFFILIATES DISCLAIM ALL LIABILITY FOR ANY USE OF THIS PRODUCT IN A WAY THAT MAY CAUSE ACCIDENTS, DAMAGE OR THAT MAY VIOLATE THE LAW.

Governing Language: This statement, any instruction manuals, user guides and other information relating to the product (Documentation) may be translated to, or has been translated from, another language (Translation). In the event of any conflict between any Translation of the Documentation, the English language version of the Documentation will be the official version of the Documentation.

This manual represents the product as at the time of printing. Navico Holding AS and its subsidiaries, branches and affiliates reserve the right to make changes to specifications without notice.

Compliance

When IS40 PRO is installed in a network with RF25N or RF70N as Rudder Feedback unit, the system complies with the following regulations:

- Wheelmark (Marine Equipment Directive 96/98 EC as amended)
- CE (2004-108 EC EMC Directive)
- C - Tick

For more information refer to our website:

pro.simrad-yachting.com

The Wheelmark

The IS40 PRO indicator system is produced and tested in accordance with the European Marine Equipment Directive 96/98. This means that the systems comply with the highest level of tests for nonmilitary marine electronic navigation equipment existing today.

The Marine Equipment Directive 96/98/EC (MED), as amended by 98/95/EC for ships flying EU or EFTA flags, applies to all new ships, to existing ships not previously carrying such equipment, and to ships having their equipment replaced.

This means that all system components covered by annex A1 must be type-approved accordingly and must carry the Wheelmark, which is a symbol of conformity with the Marine Equipment Directive.

Navico has no responsibility for incorrect installation or use of the system, so it is essential for the person in charge of the installation to be familiar with the relevant requirements as well as with the contents of the manuals, which covers correct installation and use.

Copyright

Warranty

The warranty card is supplied as a separate document.

In case of any queries, refer to our website:

pro.simrad-yachting.com

About this manual

This manual is a reference guide for operating the Simrad IS40 PRO indicator system. The manual will be continuously updated to match new sw releases. The latest available manual version can be downloaded from our web sites.

Important text that requires special attention from the reader is emphasized as follows:

→ ***Note:*** Used to draw the reader's attention to a comment or some important information.

> ⚠ Warning: Used when it is necessary to warn personnel that they should proceed carefully to prevent risk of injury and/or damage to equipment/personnel.

Contents

1

Basic Operation

Simrad IS40 PRO is a networked instrument system showing Rudder Angle in the range ±45° or ±70° (setup alternative). The display can also be configured to show up to six additional pages for speed, depth, heading, position, navigational data etc when available on the network.

→ ***Note:*** Only the Rudder Angle page is Wheelmark approved.

The IS40 PRO Display

1 Enter/Backlight key

Used to enter the main menu, select sub menus and confirm selection.

Hold 3 seconds for illumination setting.

2 Page key

If more pages than the Rudder Angle page have been enabled - scrolls through the pages.

When in menu operation - navigates one step back.

When in calibration menu - moves digit cursor left/right.

3 Screen refresh indicator

Alternating dots when instrument is alive.

4 Arrow keys

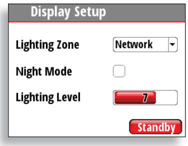
Scroll up and down through selected menus/set values.

Turning the IS40 PRO on and off

The IS40 PRO has no **Power** key, and it will be running as long as the power is connected.

Display Lighting

Press and hold the **Enter** key for 3 seconds to display the Display Setup dialog.



The Lighting Level option is active by default, and you need to press the **Enter** key to select the other options in the dialog.

1. Use the **Arrow** keys to select an option.
2. Press the **Enter** key to confirm the selection. The frame around the option is red by default and will turn green when selected to adjust values.
3. Use the **Arrow** keys to adjust the value.
4. Confirm the selection by pressing the **Enter** key.

Lighting zone

Set the lighting zone for the display. All units in the selected Lighting Zone will mirror each others light settings. Default setting is Network.

Night mode

Change the display to Night Mode color pallet. All displays in the selected Lighting Zone will also change to Night Mode.

Lighting level

Adjust the backlight level from 0-10.

→ ***Note:*** When set to level 0, the display backlight will be off and keys backlight will be at a minimum.

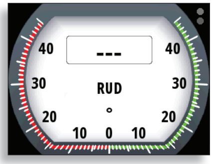
When the IS40 PRO is powered on, the Lighting level will be as set before the unit was powered off. The Night Mode option will be on if it was enabled when the unit was powered off.

Standby

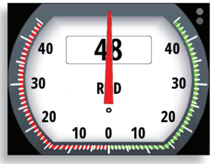
Turns off the backlight for keys and display. Press any key to return to normal operation.

Missing or faulty data

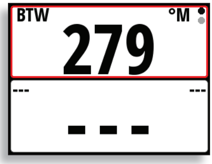
If a data type is missing or if the data is out of scale, there will be no data reading on the display.



Rudder Angle page with missing data



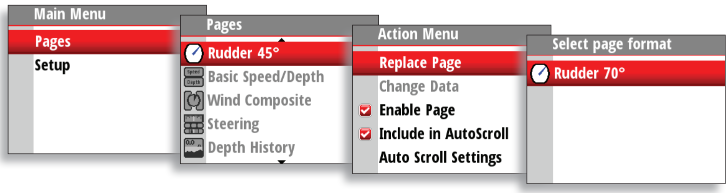
Rudder Angle page with data out of scale.



Digital page with missing data

Setting up the Rudder Angle page

Factory default analog scale for Rudder Angle is ±45°. It can be replaced by ±70° scale as shown below.



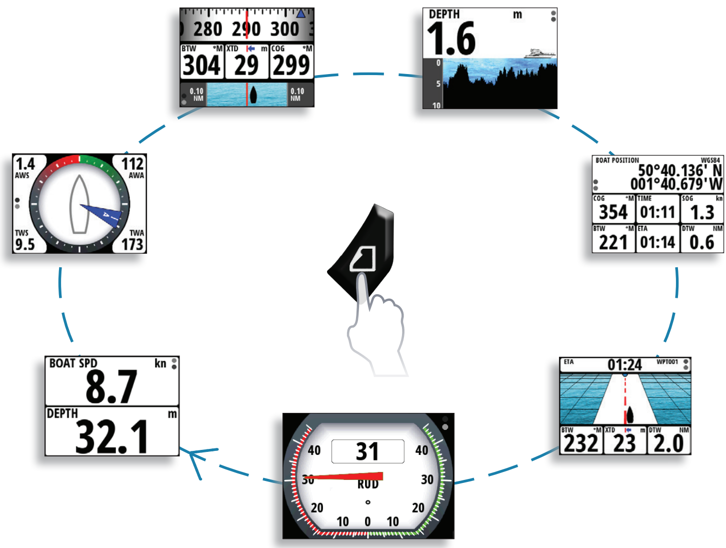
Optional data pages

From factory the display shows Rudder Angle page only. Six additional data pages may be enabled to show a variety of boat data and information available from sensors and devices on the network.

→ ***Note:*** Only the Rudder Angle pages are Wheelmark approved.

Below is an example with six additional pages enabled.
Each press of the **Page** key will change the current data page to the next preselected page in the cycle.

→ ***Note:*** If a data page has been shown for more than 5 seconds, first press on the **Page** key will show Rudder Angle page.



→ ***Note:*** Two or more pages need to be enabled for the **Page** key to function.

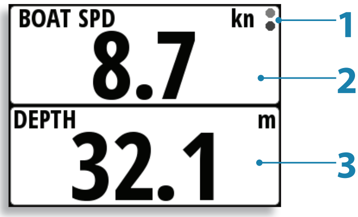
Predefined pages

→ ***Note:*** These predefined pages are not Wheelmark approved.

Basic Speed / Depth

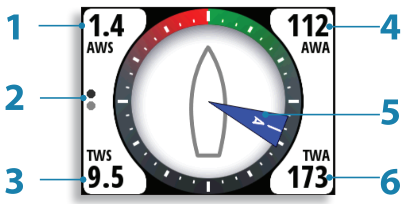
Two line data display. Boat speed and Depth

1. Refresh indicator
2. Boat speed
3. Depth



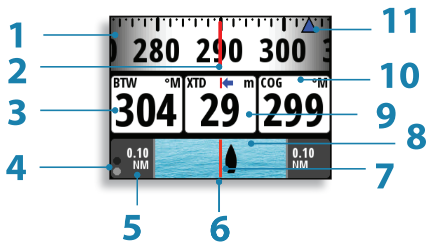
Wind

1. Apparent wind speed (AWS)
2. Refresh indicator
3. True wind speed (TWS)
4. Apparent wind angle (digital) (AWA)
5. Apparent wind angle (analog)
6. True wind angle (TWA)



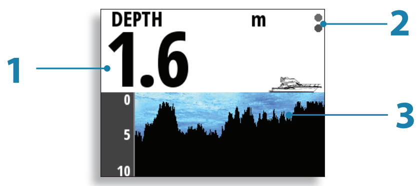
Steering

1. Compass graphic (Heading)
2. Heading
3. Bearing to waypoint (BTW)
4. Refresh indicator
5. Off track limit
6. Track
7. Boat position from track
8. Cross track distance graphic
9. Cross track distance (XTD)
10. Course over ground (COG)
11. Bearing to waypoint indicator



Depth History

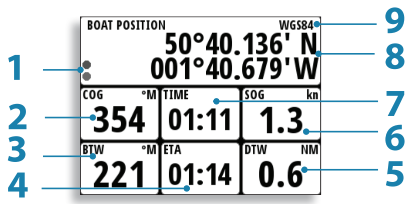
1. Depth value
2. Refresh indicator
3. Depth graphic



→ ***Note:*** You can adjust the time period scale via the **Arrow** keys. (Default = 10 min)

GPS

1. Refresh indicator
2. Course over ground (COG)
3. Bearing to waypoint (BTW)
4. Estimated time of arrival (ETA)
5. Distance to waypoint (DTW)
6. Speed over ground (SOG)
7. Local time
8. Boat position (Latitude & Longitude)
9. Coordinate system



→ ***Note:*** GPS information relies on a suitable GPS connected to the network and selected on the display as the current Position source.

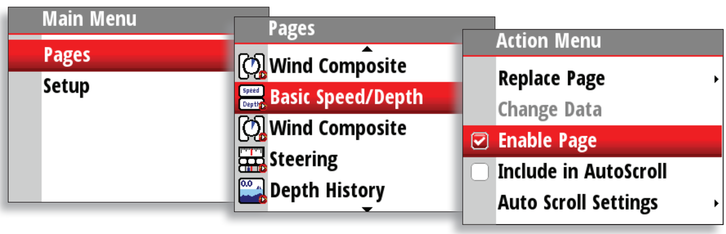
Highway

1. Estimated time of arrival (ETA)
2. Bearing to waypoint (BTW)
3. Cross track distance (XTD)
4. Distance to waypoint (DTW)
5. Highway graphic
6. Next waypoint
7. Refresh indicator
8. Waypoint name

Enabling data pages

To make a data page available via the **Page** key you will need to first ensure it has been selected as one of the seven available pages.

Once the page has been selected as one of the seven data pages you can enable it as shown below:

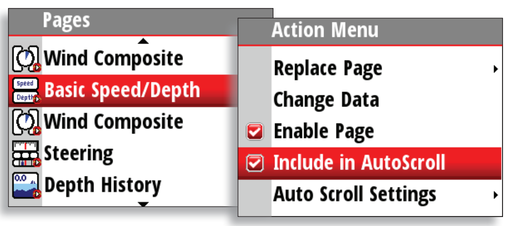


Auto scroll

When selected, auto scroll automatically scrolls between the enabled pages at a timed interval predetermined by setting the desired scroll time in the auto scroll settings menu.

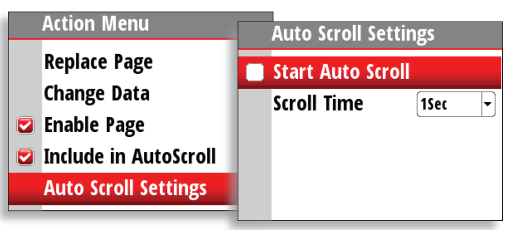
Include in auto scroll

To include a page in auto scroll, go to the auto scroll settings in the action menu of the specific page and select Include in auto scroll. Once selected a tick will appear in the check box.



Auto scroll settings

In the auto scroll settings menu you can start the auto scroll function and set the time interval between page changes.

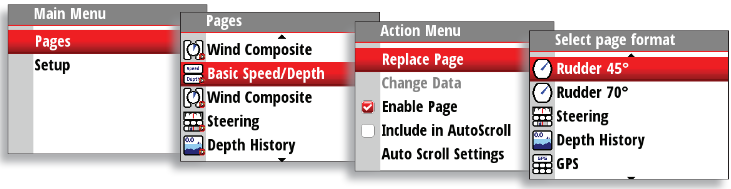


→ ***Note:*** The scroll time interval can be set to change the displayed data page between 1 and 10 second intervals.

Replacing a data page

Go to the pages menu. Select the page you wish to replace then select the new page you would like to replace it with.

→ ***Note:*** Rudder 45° page can only be replaced by Rudder 70° page.



Customizing a page

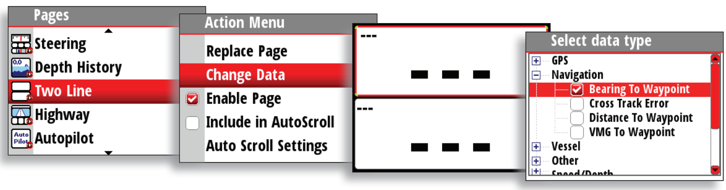
The content of the predefined pages can be changed when the page is enabled. You can replace the page with another predefined page, or you can use one of the template pages shown below.

Single data item	Histogram with a data value
Two data items	Analog display
Four data items	Full screen analog display
Four data items	Highway graphic
Nine data items	

1. Select Pages from the main menu.
2. Select one of the predefined pages, then enable the page.

3. Select Replace Page.
4. Select one of the predefined pages to replace the page, or one of the template pages if you want to create a customized page.
5. If you have used a template, select Change data and press the **Enter** key to fill a field with data from the menu. Once selected, a tick will appear in the check box. To populate other blank fields, press the **Page** key, select the field using the **Arrow** keys and press the **Enter** key to add data to the field.
6. Press the **Page** key to view the display page, and press it once more to set it as the prefered display page.

The image below shows how to change a page when using the Two Line template.

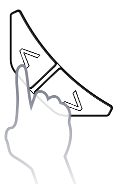


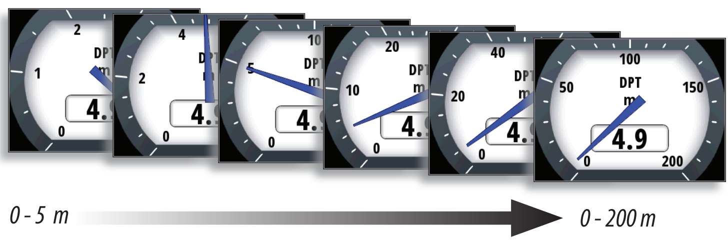
Changing an analog display scale

For some full screen analog displays (not Rudder Angle) pressing the **Arrow** keys will change the analog scale range. Select the scale range to suit your environment and requirements.

→ ***Note:*** If the actual recorded data is greater than the selected analog scale, the analog needle will remain at the highest point on the scale. The digital window in the center of the display will show the actual value.

The example below shows the available scale range for the depth analog set to meters. Pressing the **Up Arrow** key scrolls through the available analog scales from 0-5 m through to 0 -200 m. Pressing the **Down Arrow** key will decrease the analog scale.





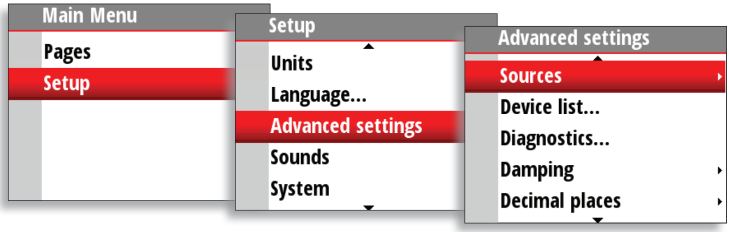
2 Setup

Sources

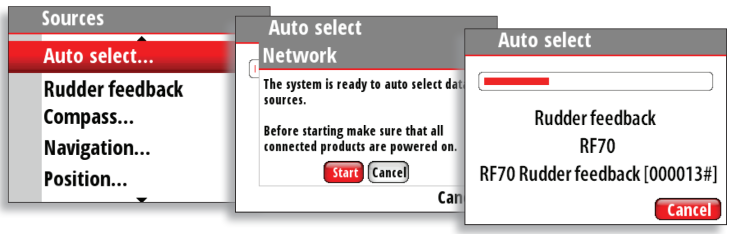
A data source can be a sensor or a device connected to the network, providing information and commands to other networked devices. The data sources are normally configured at first time turn on. It should only be necessary to update this data if a new source is added, source is missing (sensor failure), source has been enabled/ disabled, sensor replaced or a network reset.

Auto select

The Auto select option will look for all sources connected to the instrument system. If more than one source is available for each item, the display will automatically select from the internal device priority list.



1. Verify that all interfaced units are powered on.
2. Press the **Enter** key to start the auto select procedure.

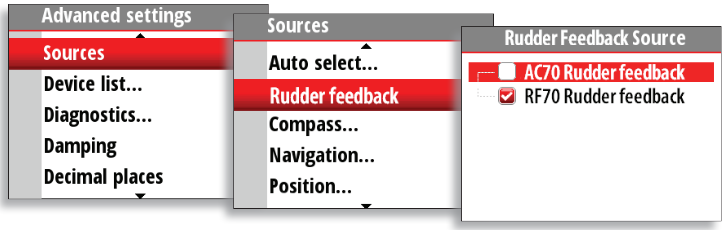


The operator will be noted when the auto select process is completed.

→ *Note:* If more than one source is available on the network, you can chose your preferred source from the sources menu. See Manual source selection.

Manual source selection

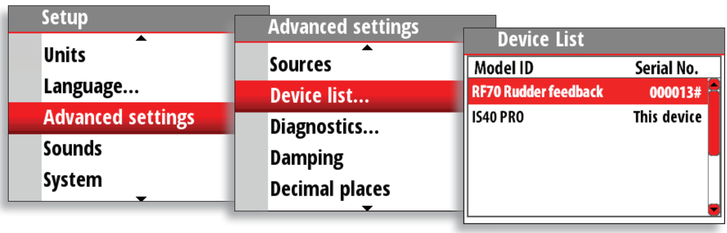
If more than one source is available for an item, the preferred source may be selected manually. As an example, the following illustrations show how the rudder feedback source is changed.



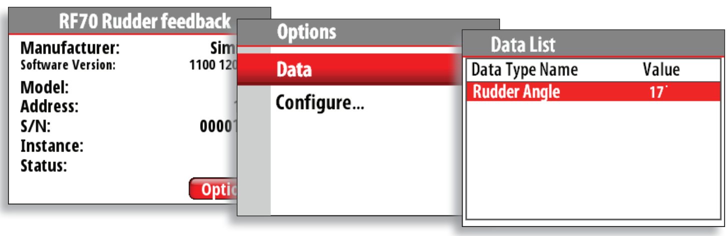
Select the preferred data source. The selected source will be indicated by a tick in the check box.

Device list

Shows a list of devices connected to the Network.



Selecting a device from the list will show you an information pane with details of that device.



Some devices, such as RF70 and RC42 compass, store their configuration, calibration and offset data in their own memory and not in the instrument display memory. For devices of this type you can check the data information, configure and calibrate the device by selecting Options.

Data

The data list shows the data type that the device is transmitting.

Configure

Instance

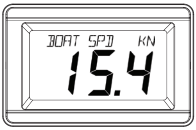
Enter a number to differentiate between instances of the same device.

Offset

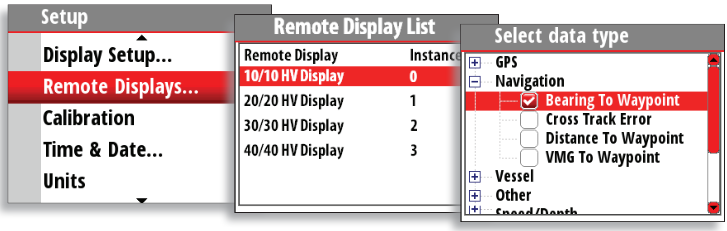
Certain devices will let you enter an offset value to compensate for the position of the sensor or variation of sensor data.

→ ***Note:*** Some devices can be configured further. If a device transmits other data it may be shown on this page also.

Remote Displays



When B&G HV displays are available on the network, they can be configured from the IS40 PRO. Follow the instructions on the screen to get the desired data.



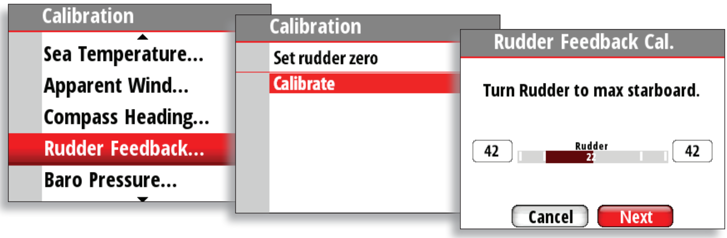
Calibration

Sensors that can be calibrated are listed in the calibration menu. For magnetic Compass Heading sensors like RC42, calibration is required. For calibration, select Calibration and follow the instructions on the display.

Rudder Feedback

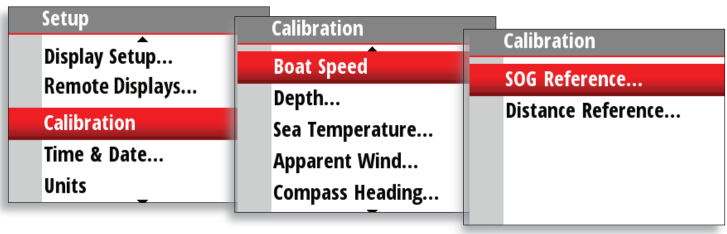
The rudder feedback sensors RF70N and RF25N are calibrated from factory. If there is a 1:1 relation between mechanical rotation of rudder stock and rudder feedback shaft, and pointer deflection is to the correct side, only zero alignment is required. If calibration is required, start calibration as shown below and follow the sequence on screen.

→ ***Note:*** When entering angle values, **Page** key is used for left/right digit selection.



Auto - Boat speed calibration via reference to GPS SOG value

This is an AutoCal facility that uses speed over ground (SOG) from your GPS and compares the average of SOG against the average boat speed from the speed sensor for the duration of the calibration run.



→ ***Note:*** This calibration should be made in calm sea with no effect from wind or tidal current.

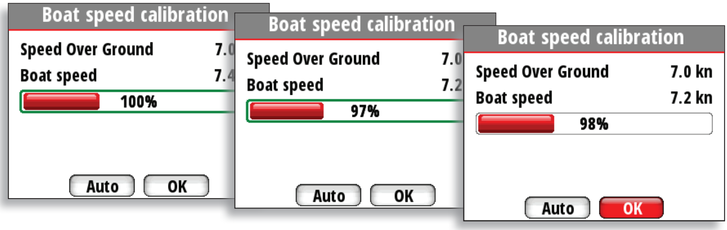
1. Bring the boat up to cruising speed (above 5 knots).
2. Select Auto on the Boat speed calibration page.
3. When the calibration is completed the Boat speed calibration scale will show the adjusted percentage value of the boat speed.

Use SOG as boat speed

If boat speed is not available from a paddle wheel sensor, it is possible to use speed over ground from a GPS. SOG will be displayed as boat speed and used also in the true wind calculations and the speed log.

Manual adjustment of boat speed

Adjust the boat speed manually by selecting the Boat speed percentage slider. Adjust the percentage up or down as desired. Confirm the value. Select OK once complete.



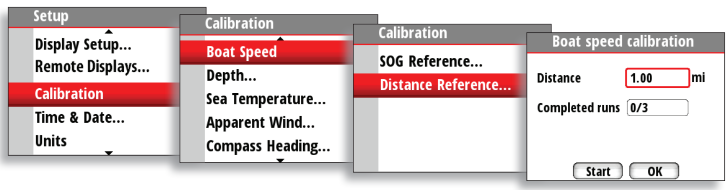
Distance Reference

This option enables the user to calibrate the log accurately and simply. Calculations are performed by the display that works out the boat speed over a known distance.

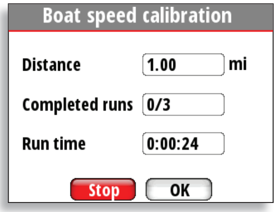
To calibrate the boat speed via a distance reference you will need to complete consecutive runs, under power at a constant speed made along a given course and distance.

→ ***Note:*** To eliminate the effect of tidal conditions it is advisable to perform at least two runs, preferably three, along the measured course.

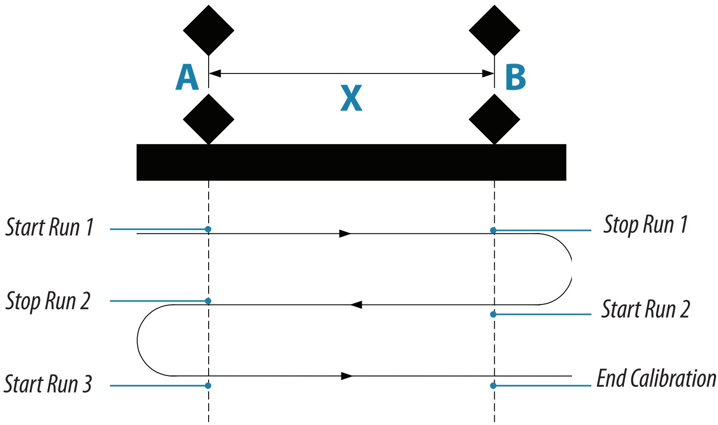
Enter the desired distance in nautical miles that you would like to calculate the distance reference over.



When the boat gets to the predetermined starting position of the distance reference calculation, start the calibration timer.

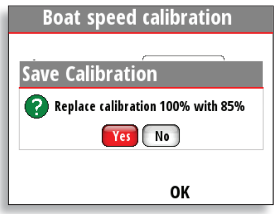


Referring to the diagram, A and B are the markers for each run and X is the actual distance for each run as measured from a suitable chart.



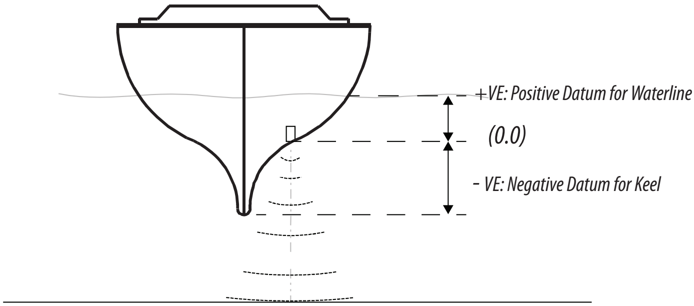
As the boat passes marks A and B on each run, instruct the system to start (Start Run) and stop (Stop Run) and finally OK to end calibration (End Cal Runs).

After the last run is completed and OK has been selected, a pop-up warning will ask you if you wish to replace the current calibration with the new one. Select Yes to complete.



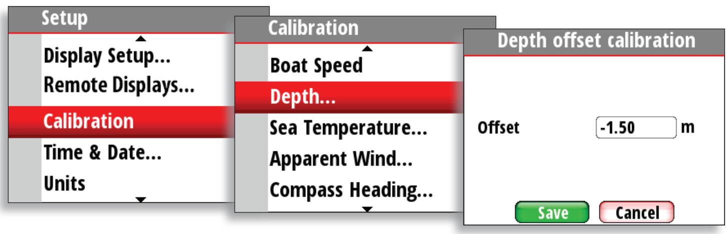
Depth

A datum (offset value) can be set, such that the depth display refers to either the water line or the base of the keel.



Setting the depth offset displays depth readings from directly below the keel or propellers of the boat, or from the waterline to the seabed. This makes it easier to see the available depth, taking into account the draught of the boat.

The offset value to be entered should represent the distance between the face of the depth transducer, and the lowest part of the boat below the waterline, or the distance between the face of the depth transducer and the water surface.



Sea Temperature

If a suitable temperature sensor is fitted, the system will monitor the current sea temperature.

The offset value to be entered should adjust the temperature reading from the sensor to match a calibrated thermometer when submersed in the water.

Apparent Wind

This provides an offset calibration in degrees to compensate for any mechanical misalignment between the masthead unit and the center line of the vessel.

Set to positive (+) reading gives offset to Starboard. Set to negative (-) reading gives offset to Port.

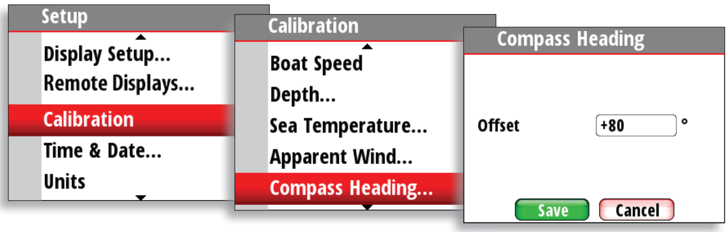
Compass Heading

The compass offset compensates for fixed errors (misalignment) between the compass sensor and the direction of the boat.

To accurately enter a compass offset, the boat's heading must be referenced to, for example: a calibrated bowl compass.

The offset value will be the difference between the known source and the currently displayed heading.

Enter this value as the offset in the compass heading field as a plus or minus integer up to 180º.



Use COG as heading

If heading data is not available from a compass sensor it is possible to use course over ground from a GPS. COG will be displayed as heading and used in the calculation of true wind direction.

Time & Date

From the time and date menu you can set your preferred time/date format and local time offset.

→ ***Note:*** Local time is calculated based on UTC provided via a GPS unit connected to the network.

Units

Set the preferred unit of measurement you want data to be displayed in.

Language

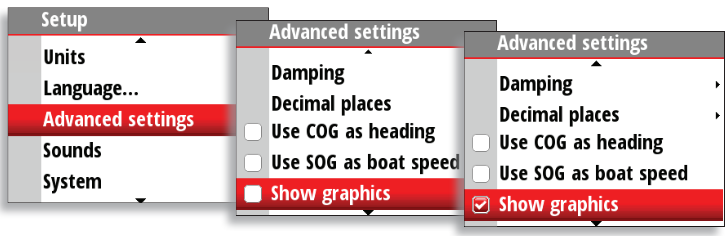
The display can be set to different languages to suit your preference.

Display setup

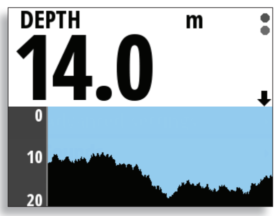
Set the Lighting Zone, enter Night Mode and change the Lighting Level. See "Display Lighting" on page 5.

Show graphics

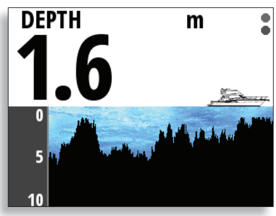
It is possible to turn on or off background graphics for some pages. Example shown below.



Background graphics off



Background graphics on



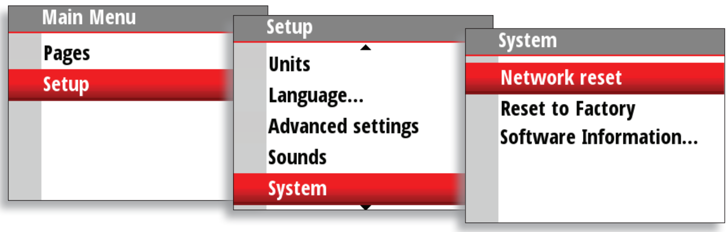
→ ***Note:*** Graphics cannot be individually set on or off for each page.

Damping
The damping rate effects the frequency that the sensor data is updated on the display, the greater the damping value the smoother the number change will be but the slower the response will be to data change.

Decimal places
It is possible to change how many decimal places (1 or 2) speed and sea temperature data will be displayed with.

Sounds
Turn the keypress sound on or off.

System
From the system menu there are several options to reset the system, and get the current software information.



Network reset
Resets the source selection on all displays connected to the network.

Reset to Factory
Resets the current display to the default settings. When the unit is restarted you will see the original startup wizard asking you to set the display.

Software information
Reveals the software you are currently running. See more information on "Software information and upgrade" on page 25.

Diagnostics

Shows an overview of the data being transmitted on the network. The list shows the network bus status, bus load as a percentage as well as quantity and type of data messages.

→ ***Note:*** We recommend that you use this diagnostic tool as a basic overview of the network status. For more detailed information it is suggested that you check the individual source information via the device list.

3

Maintenance

General maintenance

The instruments are repair-by-replacement units, and the operator is therefore required to perform only a very limited amount of preventive maintenance.

If the unit requires any form of cleaning, use fresh water and a mild soap solution (not a detergent). It is important to avoid using chemical cleaners and hydrocarbons such as diesel, petrol etc.

Always put on the weather cover when the unit is not in use.

Checking the keys

Make sure that no keys are stuck in the down position.

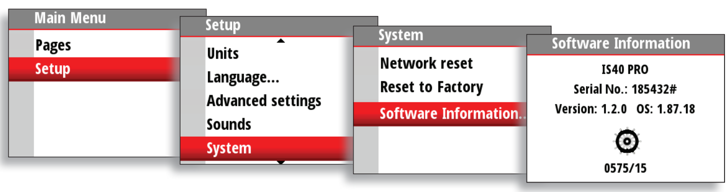
Checking the connectors

The connectors should be checked by visual inspection only. Ensure that cables are connected correctly and any unused terminals are protected.

Software information and upgrade

To find out the latest version of software available for your display, go to the Simrad website pro.simrad-yachting.com

To verify what software you are currently running, go to the software information page on your display.

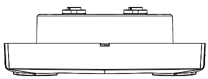


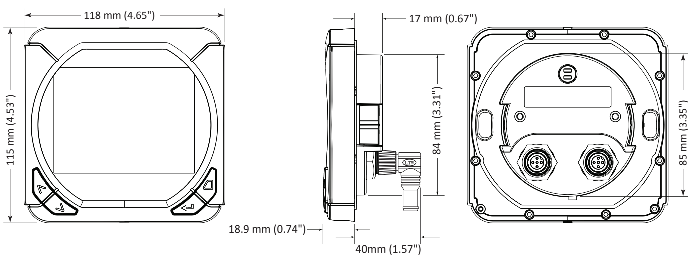
4 Specifications

Technical specifications

Weight		0.28 kg (0.6 lbs)
Power consumption		155 mA at 13.5 V
Network load		3 LEN (Load Equivalency Number)
Color		Black
Display		
	Size	4.1" (Diagonal) 4:3 Aspect ratio
	Type	Transmissive TFT-LCD - White LED backlight
	Resolution	320 x 240 pixels
	Illumination	White (day mode) / Red (night mode)
Environmental Protection		IEC 60945 protected (IPX5)
Safe distance to compass		0.2 m (0.6 ft)
Temperature		-25 to +55 ºC (-13 to +131 ºF)

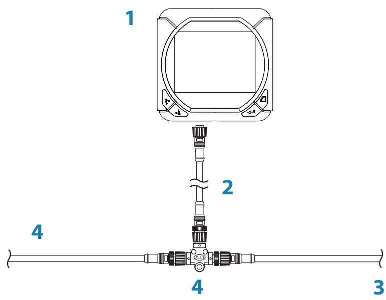
Dimensions





Wiring

1. IS40 PRO Display
2. Drop cable
3. CAN bus backbone
4. Mico-C T-Connectors



Menu flow chart

Main Menu	Level 2	Level 3	Level 4
Pages	Rudder 45°/70°		
	(Basic speed / depth)		
	(Wind composite)	Replace page	Select page format
	(Steering)	Enable page	
	(Depth history)	Include in auto scroll	
	(GPS)	Auto scroll settings	Start auto scroll
	(Highway)		Scroll time
Setup	Display setup	Lighting zone	Select group
		Night mode	
		Lighting level	
		Standby	
	Remote Displays	[List]	
	Calibration	Rudder Feedback	SOG reference
			Distance reference
		[List]	

Main Menu	Level 2	Level 3	Level 4
Setup	Time & Date	Date format	
		Time format	
		Local time	
	Units	[List]	
	Language	[List]	
	Advanced settings	Sources	[List]
		Device list	Model ID / Serial No.
		Diagnostics	[List]
		Damping	[List]
		Decimal places	Speed
			Sea temp
		Use COG as heading	
		Use SOG as boat speed	
		Show Graphics	
	Sounds	Keypress sound	
	System	Network reset	
		Reset to factory	
		Software Information	

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